

# BioSpecInfo

## Software Architecture Document (SAD) — BioSpecInfo

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# Software Architecture Document (SAD) — BioSpecInfo

| Field          | Value                                                                   |
|----------------|-------------------------------------------------------------------------|
| Project        | BioSpecInfo                                                             |
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| Thesis advisor | Prof. Savino Longo — University of Bari Aldo Moro                       |
| Type           | Progressive Web App (PWA), client-side, local-first, offline-first      |
| Platform       | Modern browsers (Chromium, Firefox, Safari/WebKit) · Desktop and Mobile |
| Repository     | <code>samupropiol-ship-it/BioSpecInfo-v11</code>                        |

## 1. System overview

BioSpecInfo is a **monolithic yet modular** web platform that runs all of its application logic inside the client browser, with no application server. It is distributed as an installable **Progressive Web App** ( `display: standalone` ) and works **offline** thanks to a Service Worker that precaches the heavy assets.

**Architectural pattern:** Single-Page/Single-File Application + satellite sub-apps in iframes, with a WebAssembly-based cheminformatics *core* decoupled from the presentation logic.

### 1.1 Guiding principles

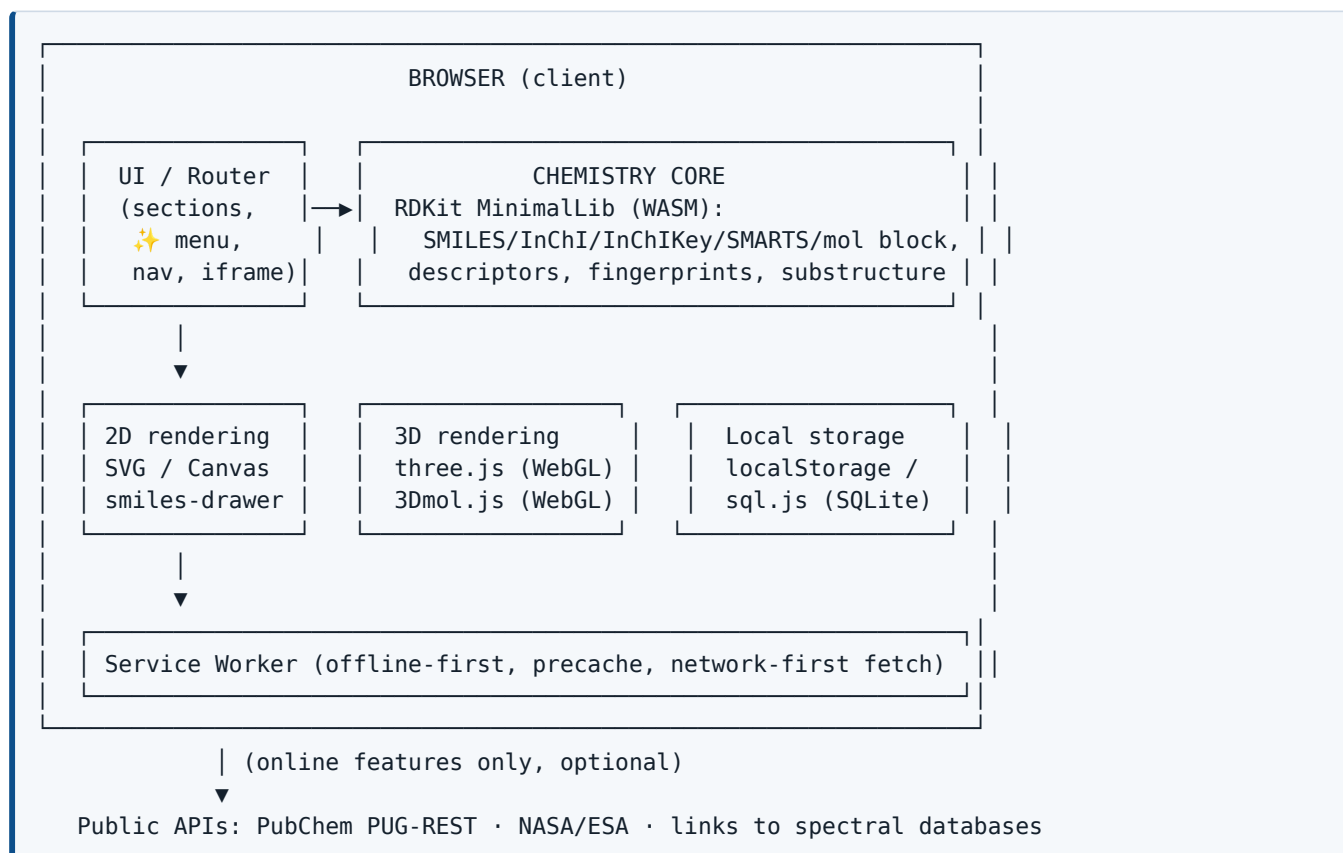
- **Local-First:** data and computations live on the device; no mandatory transmission to third parties.
- **Zero-backend:** no server database, no dynamic hosting cost (deployable on static hosting, e.g. GitHub Pages).
- **Progressive enhancement:** online features (PubChem search, real 3D models, astronomical imagery) degrade gracefully when offline.

## 2. File structure

| File                       | Role                                                                                                 |
|----------------------------|------------------------------------------------------------------------------------------------------|
| <code>index.html</code>    | Main application (chemistry core, Spectra Centre, Data Science, ✨ menu). ~4.5 MB, over 43,000 lines. |
| <code>chimorga.html</code> | Organic Chemistry course (25 modules).                                                               |
| <code>astro.html</code>    | 3D Astrochemistry (three.js).                                                                        |

| File                                                                                                                                                                                                     | Role                                                                                                                                          |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| pro.html, rdkit_lab.html, accademia.html, simulazioni.html, guidaret.html, file_manager.html, Biochimica_Guida_Definitiva.html, sr_completo.html, sr_essenziale.html, download.html, changelog_tesi.html | Thematic sub-apps loaded in iframes with <code>?v=timestamp</code> cache-busting.                                                             |
| manifest.json                                                                                                                                                                                            | PWA configuration (icons, theme, standalone).                                                                                                 |
| sw.js                                                                                                                                                                                                    | Service Worker: <i>network-first</i> strategy with cache fallback; precache of heavy libraries/assets; cache version bumped on every release. |
| RDKit_minimal.js / RDKit_minimal.wasm                                                                                                                                                                    | Cheminformatics engine (RDKit MinimalLib).                                                                                                    |
| 3Dmol-min.js                                                                                                                                                                                             | 3D viewer for molecules/macromolecules.                                                                                                       |
| three.min.js, three_bloom.js, gltf_loader.js                                                                                                                                                             | WebGL 3D rendering (anatomy, astrochemistry, mechanisms).                                                                                     |
| smiles-drawer.min.js                                                                                                                                                                                     | 2D depiction of SMILES.                                                                                                                       |
| lib/sql-wasm.js / lib/sql-wasm.wasm                                                                                                                                                                      | SQLite in WebAssembly (SQL playground).                                                                                                       |
| models/, textures/                                                                                                                                                                                       | 3D assets (organs, écorché, planets) and textures.                                                                                            |

### 3. Component architecture



### 3.1 Module Router / UI

Navigation between sections via CSS classes ( `.section.on` ) and a ✨ menu (FAB) that opens the special tools (Spectra Centre, Retrosynthesis, Quantum Lab, Data Science, etc.). Sub-apps are isolated in versioned `iframes`.

### 3.2 WebAssembly Engine (RDKit MinimalLib)

Loaded on demand ( `window.bsiLoadRDKit` ) with `locateFile` pointing to the local `.wasm` files and a CDN as a last-resort fallback. It exposes:

- **Identifiers:** canonical/isomeric SMILES, CXSMILES, InChI, InChIKey, SMARTS, MDL V2000 mol block.

- **Descriptors:** molecular weight, monoisotopic exact mass, LogP (Wildman-Crippen), TPSA, HBD/HBA, rotatable bonds, aromatic/aliphatic rings, Fsp<sup>3</sup>, molar refractivity, stereocentres.

- **Fingerprints & similarity:** Morgan ECFP4 (1024 bit) and the Tanimoto index.
- **Substructure matching** via SMARTS ( `get_qmol` + `get_substruct_matches` ).
- **2D depiction** ( `get_svg` , `get_svg_with_highlights` ) and 2D coordinate generation (mol block) re-importable into the editor.

### 3.3 Rendering

- **2D:** proprietary SVG (reaction mechanisms with Bézier arrows), Canvas 2D (molecule editor, CIP stereochemistry viewer), smiles-drawer.
- **3D:** `three.js` (WebGL) for anatomy (Drug Atlas), astrochemistry and 3D mechanisms; `3Dmol.js` for molecular models from PubChem SDF.

### 3.4 Storage Layer

- `localStorage` (with a **safety guard** and in-memory fallback for incognito / blocked-storage scenarios).
- `sql.js` (SQLite compiled to WASM) for the Data Science SQL playground.

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## 4. Implemented features (current state)

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### 4.1 Spectra Centre (molecular editor & analysis)

Draw from scratch (full periodic table, ring/group templates, drag-to-bond with snapping) or load via SMILES / library (63 molecules) → full analysis:

- **All chemical notations** (formula, canonical SMILES, CXSMILES, SMARTS, InChI, InChIKey, mol block, common/IUPAC name, CAS) with one-click copy.

- **Predicted spectra** as sub-panels: IR,  $^1\text{H}$ -NMR,  $^{13}\text{C}$ -NMR, MS with **\*\*isotope pattern (IUPAC abundances) and adduct masses\*\*** ( $[\text{M}+\text{H}]^+$ ,  $[\text{M}+\text{Na}]^+$ ...), UV-Vis, Raman — each with a bibliographic source.
- **Functional groups** (~30, via SMARTS) and **properties** (H donors/acceptors, aromatic, rotatable, stereocentres) **highlightable on the 2D and 3D structure**.
- **Drug-likeness / ADMET**: Lipinski, Veber, Egan, Ghose, Muegge rules; estimated ESOL solubility; QED (when exposed by the engine); **Brenk/PAINS structural alerts**; physicochemical radar profile.
- **Elemental analysis** (weight %) and **Tanimoto similarity** against the library.
- **3D models** (ball & stick, spacefill, wireframe) from PubChem; SVG/PNG/MOL export.

## 4.2 Synthesis & Retrosynthesis

46 disconnection strategies with an **animated mechanism player** in **2D and 3D**: curved *electron-pushing* arrows, reaction intermediates, and a retrosynthesis mode showing **synthon formation**.

## 4.3 Organic Chemistry ( [chimorga.html](#) )

25 modules: fundamentals, stereochemistry, alkenes/alkynes, aromatics & EAS, SN/E, carbonyls, amines, carbohydrates, amino acids, lipids, nucleotides, animated syntheses, named reactions, heterocycles, pericyclic & Woodward-Hoffmann, organometallics & cross-coupling, radicals, spectroscopy, retrosynthesis & protecting groups.

## 4.4 Data Science

3D Drug Atlas (13 body regions, 63 diseases of which 23 tumours, 3D organ models, PubMed links), interactive ML labs, SQL playground (sql.js), course roadmap.

## 4.5 Other sections

3D astrochemistry, Quantum Lab, Academy (gamified quizzes with SM-2 spaced repetition), interactive simulations, guides, databases of elements/drugs/ biomolecules.

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## 5. Architectural roadmap (NOT yet implemented)

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The following items belong to the *enterprise/HPC* vision and require external binaries, models or build pipelines that cannot be hosted inside a single-file PWA:

- HPC parallelism via Web Workers + [SharedArrayBuffer / Atomics](#) .
- WebGPU acceleration for dense/sparse algebra and GNN inference.
- ML predictors in ONNX Runtime WebGPU (GNN for NMR such as ShiftML2/DeepNMR; hERG, BBB, CYP450, LogS).

- In-browser molecular docking (AutoDock Vina/Webina, PDBQT format).
- Semi-empirical quantum chemistry (xTB/GFN2-xTB in WASM).
- Rule-based computational retrosynthesis engine (SMIRKS).
- SQLite-WASM database at the 1,000,000-structure scale and SDF batch processing

via Web Workers.

- Immersive WebXR (VR/AR) visualisation.
  - Native packaging (Android TWA, iOS, Electron/Tauri) and CI/CD pipelines.
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## 6. Requirements and constraints

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- **Browser:** WebAssembly and WebGL support; JavaScript enabled.
- **Network:** not required for the core; recommended for PubChem search and real 3D models.
- **Performance:** target 30–60 fps for interactive rendering; heavy computations are delegated to RDKit WASM and kept responsive.